

BEE 208 — Question 8: Average Current of Three-Phase Readings

Control Statement: for loop | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables line1, line2, line3, averageCurrent (double) and i (int)

Step 3: Use a for loop from i = 1 to 6

Step 4: Prompt user to enter Line 1, Line 2, and Line 3 currents for set i

Step 5: Read line1, line2, line3

Step 6: $\text{averageCurrent} = (\text{line1} + \text{line2} + \text{line3}) / 3$

Step 7: Display averageCurrent for set i

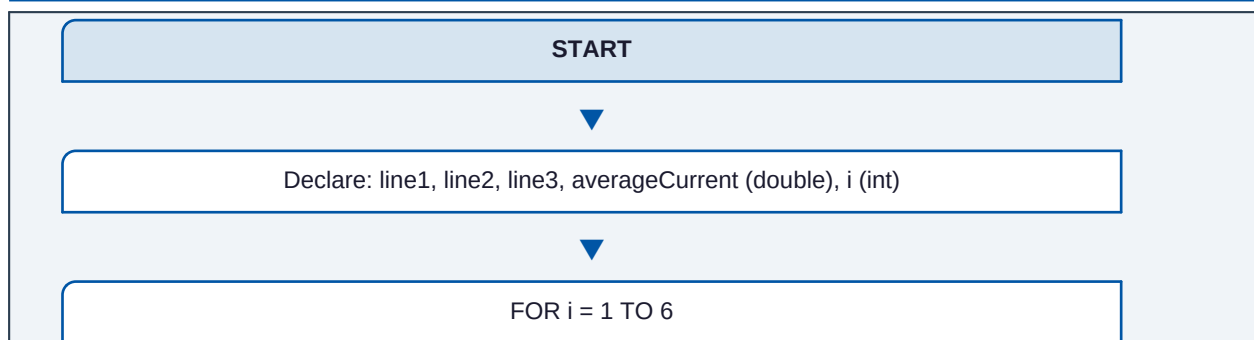
Step 8: End loop

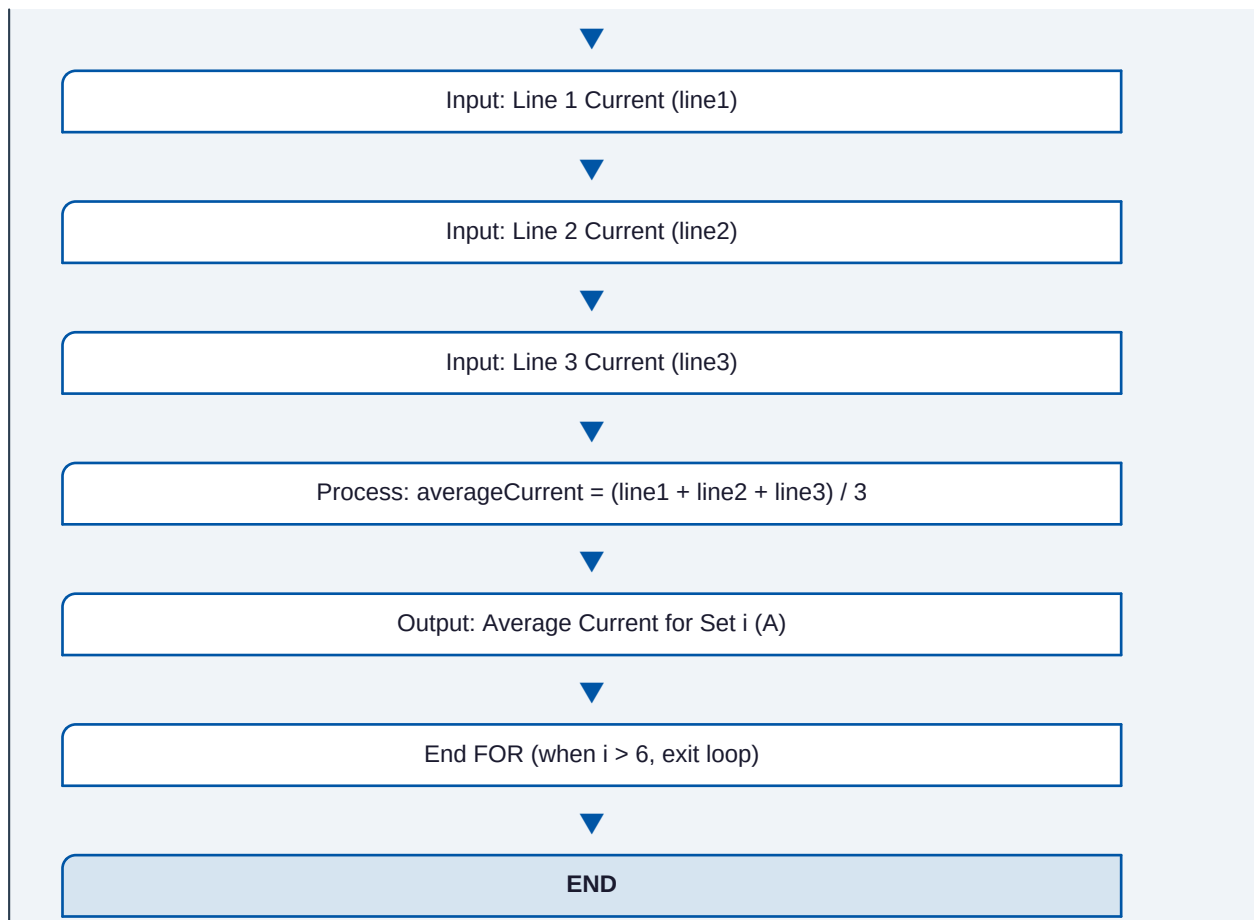
Step 9: End

2. PSEUDOCODE

```
BEGIN  
  
DECLARE line1, line2, line3, averageCurrent AS DOUBLE  
  
DECLARE i AS INTEGER  
  
FOR i = 1 TO 6 DO  
  
    OUTPUT 'Set', i  
  
    INPUT line1, line2, line3  
  
    averageCurrent = (line1 + line2 + line3) / 3  
  
    OUTPUT 'Average Current for Set', i, ': ', averageCurrent, 'A'  
  
END FOR  
  
END
```

3. FLOWCHART (Structured Diagram)





4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	Set 1: L1=10A, L2=12A, L3=11A	Expected Output	Average = 11.00 A
Input	Set 2: L1=15A, L2=15A, L3=15A	Expected Output	Average = 15.00 A